

Asset Pricing and Risk

1 Time Value of Money (Week 1)

1.1 Principal Reduction

Used when a borrower pays more than the interest due in order to reduce the loan balance.

$$\text{Remaining Principal} = \text{Initial Principal} \times (1 - \text{reduction rate})^T$$

1.2 Net Present Value

Net Present Value (NPV) is used to value projects by discounting future cash flows to time 0.

$$V_0 = CF_0 + \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \dots$$

1.3 Future Value

Future Value (FV): The value of an investment at a future date given a specific interest rate. **Future Value of an Annuity (FVA):** The accumulated value of a series of deposits at the end of a term.

$$FV = PV \cdot (1+r)^T \quad FVA(r, n) = \frac{1}{r}((1+r)^n - 1)$$

1.4 Effective Annual Rate

Effective Annual Rate (EAR): Used to compare interest rates with different compounding frequencies (n periods per year). **Implied EAR for Short-Term Loans:** A specific adaptation for loans lasting D days (e.g. payday loans).

$$EAR = \left(1 + \frac{r}{n}\right)^n - 1 \quad EAR = \left(\frac{FV}{PV}\right)^{365/D} - 1$$

1.5 Present Value and Annuities

Present Value of an Annuity (PVA): Used for fixed payments over a set duration, such as mortgages or retirement withdrawals.

$$PV = C \times ADF(r, T), \quad ADF(r, T) = \frac{1}{r} \left[1 - \frac{1}{(1+r)^T} \right]$$

International NPV (Discount-then-Convert): Values foreign projects by discounting in the local currency and then converting to the domestic currency at the spot exchange rate.

$$NPV_{\text{domestic}} = NPV_{\text{foreign}} \times \text{Spot Rate}$$

1.6 Growth and Discounting

Growing Perpetuity: Used for cash flows that continue indefinitely at a constant growth rate g .

$$V_0 = \frac{C}{r-g}$$

2 Fixed Income (Week 2)

2.1 Credit Risk and Portfolio Immunization

Probability of Repayment and Default: Computed when the recovery rate R , the promised yield r_p and the risk-free rate r_f are known.

$$p_R = \frac{\frac{1+r_f}{1+r_p} - R}{1-R} \quad p_D = 1 - p_R$$

Duration Matching: To immunize a portfolio against interest rate changes, the value-weighted durations must match the target duration.

$$V_1 D_1 + V_2 D_2 = D_{\text{target}} (V_1 + V_2)$$

Credit Spread: The yield difference between a corporate bond and a government bond of the same maturity.

$$\text{Credit Spread} = \text{Corporate Bond Yield} - \text{Government Bond Yield}$$

2.2 Bond Rates of Return

The rate of return measures the actual gain or loss over a specific holding period:

$$R = \frac{\text{Coupon Income} + \text{Price Change}}{\text{Initial Investment}} = \frac{P_1 + C - P_0}{P_0}$$

2.3 Bond Pricing and Yield to Maturity

Annual Coupon Bond: Using the annuity representation for the coupons,

$$P_0 = \sum_{t=1}^T \frac{C}{(1+y)^t} + \frac{F}{(1+y)^T} = \frac{C}{y} \left[1 - \frac{1}{(1+y)^T} \right] + \frac{F}{(1+y)^T}$$

Semi-Annual Coupons: The coupon is $C/2$, the yield is $y/2$ and the total number of periods is $2T$.

Pricing via Spot Rates (Bootstrapping):

$$P = \frac{C_1}{1+r_{0,1}} + \frac{C_2}{(1+r_{0,2})^2} + \dots + \frac{C_T + F}{(1+r_{0,T})^T}$$

2.4 Forward Rates and Arbitrage

Forward rates represent current forecasts of future spot rates. The forward rate between t_1 and t_2 , and the one-year forward rate, are

$$f_{t_1, t_2} = \left[\frac{(1+r_{0, t_2})^{t_2}}{(1+r_{0, t_1})^{t_1}} \right]^{\frac{1}{t_2-t_1}} - 1 \quad 1 + f_t = \frac{(1+r_{0, t})^t}{(1+r_{0, t-1})^{t-1}}$$

2.5 Measures of Interest-Rate Risk

Macauley Duration (D) and **Modified Duration (D_m):**

$$D = \frac{\sum_{t=1}^T t \cdot PV(CF_t)}{P} \quad D_m = \frac{D}{1+y}$$

Duration and Convexity Approximation: The first-order approximation of the percentage price change, and the second-order approximation including convexity, are

$$\frac{\Delta P}{P} \approx -D_m \Delta y \quad \frac{\Delta P}{P} \approx -D_m \Delta y + \frac{\text{Con}(\Delta y)^2}{2}$$

3 Risk Preferences and Utility (Week 3)

3.1 Basic Statistics and Wealth Calculations

Expected Pay-off and Variance of a random variable X :

$$E[X] = \sum_{i=1}^n p_i x_i \quad \text{Var}(X) = \sum_{i=1}^n p_i (x_i - E[X])^2$$

In the lecture notes, variance is denoted by ϖ^2 and, for fair lotteries, is associated with $E[\hat{\epsilon}^2]$.

Expected Final Wealth: $E[W_T] = W_0 + E[X]$

Time-Scaling of Volatility: Expected return scales linearly with time, while volatility scales with the square root of time, $\sigma_t = \sigma\sqrt{t}$.

Bernoulli-Style Lottery: For a lottery Z with outcomes H and $-L$, $\text{Var}(Z) = p(1-p)(H+L)^2$.

3.2 Expected Utility Theory

Expected Utility (Discrete):

$$V = E[U(W)] = \sum_{i=1}^n p_i U(W_i)$$

CRRA Utility Functions: The lecture notes define power utility as $U(W) = \frac{1}{\gamma} W^\gamma$. The exercise solutions use the standard CRRA form:

$$U(W) = \begin{cases} \frac{W^{1-\gamma}}{1-\gamma}, & \gamma \neq 1 \\ \ln(W), & \gamma = 1 \end{cases}$$

Linear Transformation: Expected utility is unique up to a linear monotonic transformation, $W(P) = a + bV(P)$.

3.3 Certainty Equivalent and Risk Premium

Certainty Equivalent: Defined implicitly by $U(CE) = E[U(W)]$. For CRRA utility (Question 2),

$$CE = \begin{cases} [(1-\gamma)V]^{\frac{1}{1-\gamma}}, & \gamma \neq 1 \\ e^V, & \gamma = 1 \end{cases}$$

These explicit inverse functions are not provided in the lecture notes, which define the certainty equivalent only implicitly.

Risk Premium: The Markowitz risk premium π is defined implicitly by $U(E[X] - \pi) = E[U(X)]$. For CRRA utility (Question 6),

$$\pi = E(X) - (E[X^{1-\gamma}])^{\frac{1}{1-\gamma}}$$

Measures of Risk Aversion: Pratt-Arrow absolute risk aversion and relative risk aversion:

$$R(W) = -\frac{U''(W)}{U'(W)} \quad R_r(W) = WR(W)$$

3.4 Prospect Theory

Independence Axiom:

$$P \succ Q \iff \alpha P + (1-\alpha)R \succ \alpha Q + (1-\alpha)R$$

Overall Value: $V(P) = \sum_i \pi_i v(x_i)$, where π_i are decision weights and $v(\cdot)$ is the value function.

Value Function and Probability Weighting (Gains): The solutions use $\alpha = 0.88$.

$$v(x) = x^\alpha \quad w^+(p) = \frac{p^\delta}{(p^\delta + (1-p)^\delta)^{1/\delta}}$$

4 Mean-Variance Portfolio Theory (Week 4)

4.1 Portfolio Optimization

Lagrangian Formulation: Minimize portfolio variance subject to a return constraint:

$$L(w, \lambda) = \frac{1}{2} w' V w - \lambda (w'(\bar{R} - R_{fe}) - (\bar{R}_p - R_f))$$

Relationship between λ and γ : Without a risk-free asset, the first-order conditions of the Markowitz problem give

$$\lambda = \frac{\delta \bar{R}_p - \alpha}{\Delta} \quad \gamma = \frac{\zeta - \alpha \bar{R}_p}{\Delta}$$

4.2 Basic Portfolio Statistics and the Sharpe Ratio

For a portfolio of one risky asset and a risk-free asset:

$$E[R_p] = w_{\text{risky}} E[R_{\text{risky}}] + w_{rf} R_f \quad \sigma_p = w_{\text{risky}} \sigma_{\text{risky}}$$

Overall Portfolio Weights: The weight of a stock in an overall portfolio that includes T-bills is

$$\text{Weight} = (\text{weight of risky fund}) \times (\text{weight of stock in risky fund})$$

$$\text{Sharpe Ratio: } SR = \frac{\bar{R}_i - R_f}{\sigma_i}$$

4.3 Two-Asset Portfolios

Perfect Negative Correlation ($\rho = -1$): $\sigma_P = |w_A \sigma_A - w_B \sigma_B|$

Covariance: $\sigma_{X,Y} = \rho \sigma_X \sigma_Y$, with variance $\sigma^2 = (\sigma)^2$.

4.4 Efficient Frontier and Matrix Algebra

In the n -asset case, V denotes the covariance matrix and $\bar{R}$ the vector of expected returns.

Efficient Set Constants:

$$\alpha = e' V^{-1} \bar{R}, \quad \delta = e' V^{-1} e, \quad \zeta = \bar{R}' V^{-1} \bar{R}, \quad \Delta = \zeta \delta - \alpha^2$$

Inverse of a 2×2 Covariance Matrix:

$$V^{-1} = \frac{1}{\det(V)} \begin{pmatrix} \sigma_Y^2 & -\sigma_{X,Y} \\ -\sigma_{X,Y} & \sigma_X^2 \end{pmatrix}, \quad \det(V) = \sigma_X^2 \sigma_Y^2 - (\sigma_{X,Y})^2$$

Efficient Portfolio Weights and Variance Identity:

$$w^* = \lambda V^{-1} \bar{R} + \gamma V^{-1} e \qquad \sigma_p^2 = \gamma + \lambda \bar{R}_p$$

4.5 Specific Portfolio Types

Global Minimum Variance Portfolio:

$$w_{GMV} = \frac{1}{\delta} V^{-1} e \qquad \bar{R}_{GMV} = \frac{\alpha}{\delta} \qquad \sigma_{GMV} = \frac{1}{\sqrt{\delta}}$$

Tangency Portfolio: The maximum Sharpe ratio portfolio has expected return

$$\bar{R}_{TAN} = \frac{\alpha R_f - \zeta}{\delta R_f - \alpha}$$

With a risk-free asset, the weights are

$$w^* = \lambda V^{-1} (\bar{R} - R_f e), \qquad \lambda = \frac{\bar{R}_p - R_f}{\zeta - 2\alpha R_f + \delta R_f^2}$$

4.6 Utility and Efficiency Properties

Markowitz Utility: $U(R_p) = E[R_p] - \frac{\gamma}{2} \text{Var}(R_p)$

Optimal Portfolio under Risk Aversion ρ :

$$\lambda = \frac{1}{\rho} \qquad \gamma = \frac{1}{\delta} \left(1 - \frac{\alpha}{\rho} \right)$$

Covariance with the GMV Portfolio: For any efficient portfolio,

$$\text{Cov}(R_{mv}, R_p) = \text{Var}(R_{mv}).$$

Correlation and Efficiency: For an efficient portfolio a and any portfolio b ,

$$\text{Corr}(R_a, R_b) = \frac{SR_b}{SR_a}.$$

5 Capital Asset Pricing Model (Week 5)

5.1 CAPM and Beta

The CAPM (Security Market Line) gives the fair or expected return of asset k as

$$E(r_k) = r_f + \beta_k (E[r_m] - r_f)$$

Asset Beta: Measures sensitivity to market returns.

$$\beta_k = \frac{\text{Cov}(r_k, r_m)}{\text{Var}(r_m)} = \frac{\sigma_{km}}{\sigma_m^2} = \rho_{km} \frac{\sigma_k}{\sigma_m}$$

Total Risk Decomposition: Total variance is the sum of systematic and idiosyncratic components, $\sigma_k^2 = \beta_k^2 \sigma_m^2 + \sigma_{\epsilon k}^2$.

5.2 Efficient Portfolios and the Capital Market Line

Capital Market Line: Relates expected return to total risk for efficient portfolios.

$$R_p = R_f + \frac{R_m - R_f}{\sigma_m} \sigma_p$$

Large Portfolios:

$$\sigma_p^2 = \frac{1}{n} \times \text{Average Variance} + \frac{n-1}{n} \times \text{Average Covariance}$$

Portfolio Beta: The weighted average of individual asset betas, $\beta_p = \sum_i w_i \beta_i$.

5.3 Performance Measures

Alpha measures abnormal performance relative to CAPM, and the **Treynor**

Ratio evaluates excess return per unit of systematic risk:

$$\alpha_k = \text{Actual Return} - [r_f + \beta_k (R_m - r_f)] \qquad \text{Treynor} = \frac{\mu_A - r_f}{\beta_A}$$

5.4 Project Valuation

$$NPV = \sum_{t=1}^n \frac{CF_t}{(1+r)^t} - \text{Initial Investment}$$

The present value of an annuity (PVA) is used as a numerical shortcut for constant cash flows in NPV calculations.

5.5 Scenario-Based Statistics

When outcomes are given by discrete probability states:

$$\mu = \sum_i P_i R_i \qquad \sigma^2 = \sum_i P_i (R_i - \mu)^2$$

$$\text{Cov}(X, M) = \sum_i P_i (R_{X,i} - \mu_X) (R_{M,i} - \mu_M) \qquad \rho_{AB} = \frac{\text{Cov}(r_A, r_B)}{\sigma_A \sigma_B}$$

5.6 Utility Maximization

The Lagrangian used to derive first-order conditions for investor utility maximization is

$$L = \sum_i w_i \mu_i - \frac{1}{2\gamma_k} \sum_i \sum_j w_i w_j \sigma_{ij} + \lambda \left(1 - \sum_i w_i \right)$$

6 Arbitrage Pricing Theory (Week 6)

6.1 APT Expected Return

$$E[R_i] = \lambda_0 + \sum_{z=1}^k b_{iz} \lambda_z \qquad E[R_p] = R_f + b_{p1} \lambda_1 + b_{p2} \lambda_2$$

- $\lambda_0 = R_f$
- b_{iz} = Sensitivity (beta) of asset i to factor z
- λ_z = Risk premium of factor z

The second expression is the two-factor case for a portfolio p .

6.2 Arbitrage Portfolio Construction

An arbitrage portfolio requires zero net investment, factor neutrality and a strictly positive expected return:

$$\sum_{i=1}^n w_i = 0, \qquad \sum_{i=1}^n w_i b_{zi} = 0 \quad \forall z, \qquad E[R_p] = \sum_{i=1}^n w_i E[R_i] > 0$$

One-Factor Arbitrage Ratio: Mispricing can be identified by comparing $\frac{E[R_i] - R_f}{\beta_i}$. If this ratio differs across well-diversified portfolios, an **arbitrage opportunity** exists.

6.3 CAPM and Beta Decomposition

Under the CAPM, and with the market beta expressed via APT factor sensitivities (β_{MFz} = market beta of factor z):

$$E[R_i] - R_f = \beta_{Mi} (E[R_M] - R_f) \qquad \beta_{MA} = b_{1A} \beta_{MF1} + b_{2A} \beta_{MF2}$$

6.4 Factor Transformation and Orthogonalization

New factors defined as linear combinations of the original factors, $g_1 = c_1 f_1 + c_2 f_2$ and $g_2 = c_3 f_1 + c_4 f_2$, transform the risk premia as

$$\begin{bmatrix} \lambda_{g1} \\ \lambda_{g2} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 \\ c_3 & c_4 \end{bmatrix} \begin{bmatrix} \lambda_{f1} \\ \lambda_{f2} \end{bmatrix}$$

Orthogonality Conditions: For orthogonal factors with unit variance,

$$c_1 c_3 + c_2 c_4 = 0, \qquad c_1^2 + c_2^2 = 1, \qquad c_3^2 + c_4^2 = 1$$

6.5 Covered Interest Parity

$$\frac{S_0}{F_0} (1 + r_0) = 1 + r_0^*$$

- S_0 = Spot exchange rate, F_0 = Forward exchange rate

- r_0 = Domestic interest rate, r_0^* = Foreign interest rate

The CAPM can be interpreted as a single-factor model in which all systematic risk is captured by market movements, whereas the APT decomposes systematic risk into multiple underlying economic factors, allowing asset returns to be explained by several distinct sources of risk.

7 State Prices and Complete Markets (Week 7)

7.1 Market Completeness and Matrix Algebra

Determinant Condition (2×2 Case): For assets A , B and states H , M , the payoff matrix X must be invertible, $\det(X) = A_H B_M - A_M B_H \neq 0$.

Rank Condition: A market is *complete* if and only if the row rank of X equals the number of states k , $\text{rank}(X) = k$.

Redundancy: If the determinant is zero, assets are redundant and one asset is a scalar multiple of another, $A = \delta B$.

7.2 Arbitrage Conditions

Definition of Arbitrage: A portfolio θ such that

$$P' \theta \leq 0 \quad \text{and} \quad X \theta \geq 0,$$

with at least one inequality holding strictly.

Positivity of State Prices: In a complete market, the market is arbitrage-free if and only if all Arrow-Debreu state prices are strictly positive, $q \gg 0$.

Arbitrage-Free Price Range: In the 2×2 case with $\det(X) > 0$,

$$\frac{B_M}{A_M} P_A > P_B > \frac{B_H}{A_H} P_A$$

7.3 Arrow-Debreu Securities and State Prices

$$q = (X')^{-1} P \qquad X^{-1} = \frac{1}{\det X} \begin{pmatrix} B_M & -B_H \\ -A_M & A_H \end{pmatrix}$$

The Arrow-Debreu price of state j is the cost of a portfolio that pays 1 in state j and 0 in all other states. A risk-free asset paying 1 in every state is priced as

$$P_{\text{risk-free}} = \sum_i q_i = \frac{1}{1 + r_f}$$

7.4 Firm Claims: Debt, Equity and CDS

A is the firm asset value and F is the face value of debt:

$$D = \min(A, F) \qquad E = \max(A - F, 0) = A - D \qquad p_A = p_D + p_E$$

$$\text{Payoff}_{\text{CDS}} = \begin{cases} F - A_C, & \text{if } A_C < F \\ 0, & \text{otherwise} \end{cases}$$

7.5 Advanced Matrix Solving

For sparse 3×3 markets, δ_k is the determinant of the 2×2 sub-matrix obtained by removing the last column and row k , for example $\delta_2 = A_1 B_3 - A_3 B_1$. The replication weights are

$$\theta_{3,C} = \frac{1}{C_3}, \qquad \theta_{1,C} = \frac{\delta_1}{\delta_3 C_3}, \qquad \theta_{2,C} = -\frac{\delta_2}{\delta_3 C_3}$$

7.6 Options and Put-Call Parity

$$\text{Price} = \sum_i (\text{Payoff}_i \times q_i) \qquad P + S = C + PV(K)$$

P is the put price, C the call price, S the asset price and $PV(K)$ the present value of the strike price.

7.7 Stochastic Discount Factor and Risk-Neutral Probabilities

With π_i the true probability of state i :

$$m_i = \frac{q_i}{\pi_i} \qquad \hat{\pi}_i = \frac{q_i}{\sum_j q_j}$$

8 Binomial Option Pricing (Week 8)

8.1 Asset Price Dynamics and Statistics

These formulas describe the behaviour of the stock price S and its log returns Z in a binomial tree.

Stock Price Dynamics: After k up-moves in n steps,

$$S_{n,k} = U^k D^{n-k} S_0, \quad U = e^u, \quad D = e^d$$

Conditional Moments of Log Returns:

$$E_n[Z_{n+1}] = pu + (1-p)d \quad \text{Var}(Z) = p(1-p)(u-d)^2$$

$$\text{StdDev}(Z) = (u-d)\sqrt{p(1-p)} \quad \text{Volatility}_n = \sqrt{n} \text{Volatility}_1$$

Unconditional Distribution:

$$\Pr(S_n = S_{n,k}) = \binom{n}{k} p^k (1-p)^{n-k}$$

8.2 Model Calibration

Assuming $p = \frac{1}{2}$ and a time step Δ , the parameters match the mean and variance, with gross risk-free return R per period:

$$u = \mu\Delta + \sigma\sqrt{\Delta}, \quad d = \mu\Delta - \sigma\sqrt{\Delta}, \quad R = e^{r\Delta}$$

8.3 Risk-Neutral Valuation and Replication

Risk-Neutral Probability and Backward Induction:

$$p^* = \frac{R-D}{U-D} \quad V_{n,k} = e^{-r\Delta} [p^* V_{n+1,k+1} + (1-p^*) V_{n+1,k}]$$

Replicating Portfolio: Shares of stock (delta) and bonds (alpha):

$$\delta_{n,k} = \frac{V_{n+1,k+1} - V_{n+1,k}}{S_{n+1,k+1} - S_{n+1,k}} \quad \alpha_{n,k} = \frac{\delta_{n,k} S_{n+1,k+1} - V_{n+1,k+1}}{B_{n+1}}$$

Self-Financing Condition:

$$\delta_n S_{n+1} - \alpha_n B_{n+1} = \delta_{n+1} S_{n+1} - \alpha_{n+1} B_{n+1}$$

8.4 Option-Specific Formulas

Intrinsic Value: Call, $IV_{n,k} = S_{n,k} - X$. Put, $IV_{n,k} = X - S_{n,k}$.

European and American Options: European, $V_{n,k} = CV_{n,k}$. American, $V_{n,k} = \max\{IV_{n,k}, CV_{n,k}\}$.

European Option Value (Expectation Method):

$$V_0 = e^{-rT} \sum_{k=0}^n \binom{n}{k} (p^*)^k (1-p^*)^{n-k} \cdot \text{Payoff}_{n,k}$$

9 Options and Forwards (Week 9)

9.1 Option Payoffs and Pricing

Payoffs at Maturity: For European options at expiration τ ,

$$\text{European Call: } C_\tau = \max(S_\tau - K, 0)$$

$$\text{European Put: } P_\tau = \max(K - S_\tau, 0)$$

$$\text{Written (Short) Call: } -\max(S_\tau - K, 0)$$

$$\text{Written (Short) Put: } -\max(K - S_\tau, 0)$$

Put-Call Parity: Standard form, and including the present value of dividends D :

$$C_0 + PV(K) = P_0 + S_0 \quad C_0 + R_f^{-\tau} X + D = P_0 + S_0$$

Forward and Futures Pricing: The fair forward price accounting for dividends or income, and the simple compounding form used in exercises:

$$F_{0\tau} = (S_0 - D)R_f^\tau \quad F_0 = [S_0 - PV(\text{Dividends})] \cdot (1+r)^T$$

Arbitrage Bounds and Strike Price Properties:

$$\text{Call lower bound: } C_0 \geq \max(S_0 - D - PV(X), 0)$$

$$\text{Strike property 1: } C(K_1) \geq C(K_2) \quad \text{if } K_1 < K_2$$

$$\text{Strike property 2: } P(K_2) - P(K_1) \leq K_2 - K_1$$

Continuously Compounded Present Value: $PV(K) = Ke^{-rT}$

9.2 Rate of Return and Break-Even Points

Basic Rate of Return:

$$\text{Rate of Return} = \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}}$$

Implied Continuous Rate of Return: Given PV and future value K over T periods,

$$PV \cdot e^R = K \quad \implies \quad R = \frac{\ln(K/PV)}{T}$$

Break-Even Points for Options:

$$\text{Put: } S_T = \text{Strike Price} - \text{Initial Net Proceeds}$$

$$\text{Call: } S_T = \text{Strike Price} + \text{Initial Net Proceeds}$$

Present Value (Simple Compounding): For non-indexed assets or discrete compounding, $PV = \frac{\text{Future Value}}{(1+r)^T}$.

Total Profit/Loss of a Position:

$$\text{Profit} = \text{Net Proceeds from Writing Options} - \text{Net Cash Outflows at Expiration}$$